

Worksheet: A Comparative Network and Sentiment Analysis of Rhythm Game Communities on YouTube

By: UG Group 59

Authors:

Ethan Bito (s4102812)

Roopkotha Chakrabarti (s4054393)

Contents

I.	Team Project Plan.....	3
II.	Weekly Timesheets.....	4
III.	Meeting Minutes	5
IV.	Individual Self-Reflection.....	6

I. Team Project Plan

a. Project Objective

The objective of this project was to investigate differences in sentiment and interaction patterns between two rhythm game fandom communities on YouTube: Project SEKAI and BanG Dream! Girls Band Party!. The project aimed to apply sentiment analysis and social network analysis to compare user discussion behaviour, community structure, and reply interaction patterns within these two fandom communities.

b. Team Roles and Responsibilities

Team Member	Responsibilities
Ethan Bito (s4102812)	YouTube data collection pipeline, preprocessing, sentiment analysis, network construction, visualisation generation, code packaging, report drafting.
Roopkotha Chakrabarti (s4054393)	Research planning, query testing, report drafting, interpretation of results, review analysis outputs, sentiment analysis, network construction.

c. Project Timeline

Week	Planned Tasks	Outcome
Week 9	Initial project planning, topic discussion, research question selection, assign responsibilities.	Final project scope and task division established.
Week 10	Data collection setup, YouTube API query testing, initial data collection and preprocessing.	Raw and cleaned datasets generated.
Week 11	Sentiment analysis, network construction, community detection, visualisation generation.	Core analysis completed.
Week 12	Report writing, worksheet completion, submission and code packaging, final reviews.	Submission files finalised.

d. Team Meetings

Week	Meeting Focus	Outcome
Week 9	Discuss assignment requirements and chosen topic.	Research question and project plan agreed.
Week 10	Review collected data and refine queries.	Final games and data collection strategy confirmed.
Week 11	Review analysis outputs and discuss interpretation.	Key findings selected for report.

Week 12	Final report review and submission checklist.	Final submission prepared.
----------------	---	----------------------------

II. Weekly Timesheets

Team Member 1 – Ethan Bito (s4102812)

Week	Task Completed	Hours
Week 9	Participated in initial planning meeting, discussed project scope and research question, assisted with data source planning and query brainstorming.	4 hours
Week 10	Developed YouTube data collection pipeline, tested search queries, collected initial data, assisted with preprocessing and review with group.	9 hours
Week 11	Performed sentiment analysis, built reply networks, generated visualisation, discussed analysis outputs and interpretation with group member.	10 hours
Week 12	Finalised code packaging, assisted with report writing, worksheet preparation and finalisation, reviewed submission files and checklist.	7 hours
		Total: 30 hours

Team Member 2 – Roopkotha Chakrabarti (s4054393)

Week	Task Completed	Hours
Week 9	Participated in initial planning meeting, contributed to research planning, discussed project scope and research question.	4 hours
Week 10	Assisted with query testing, reviewed data collection outputs, contributed to preprocessing checks and early report planning.	6 hours
Week 11	Assisted with sentiment analysis and network analysis interpretation, reviewed visualisation outputs, contributed to findings discussion and report drafting.	10 hours
Week 12	Worked on report drafting, interpretation of final results,	9 hours

	worksheet preparation, final review of report and submission materials.	
		Total: 29 hours

III. Meeting Minutes

Meeting 1 – Week 9 (Project Planning)

Attendees: Ethan Bito (s4102812), Roopkotha Chakrabarti (s4054393)

Agenda:

- Discuss assignment requirements.
- Brainstorm possible project topics.
- Decide on research question and project scope.
- Assign initial responsibilities.

Discussion/Decisions:

- Reviewed Assignment 2 requirements and rubric.
- Agreed to focus on comparative analysis of two rhythm games on YouTube.
- Selected Project SEKA! and BanG Dream! Girls Band Party! as case studies.
- Drafted initial research question focused on sentiment and reply network differences.
- Agreed on task division and project timeline.

Action Items:

- Ethan: Begin YouTube API setup and query testing.
- Roopkotha: assist with research planning and query refinement.

Meeting 2 – Week 10 (Data Collection Review)

Attendees: Ethan Bito (s4102812), Roopkotha Chakrabarti (s4054393)

Agenda:

- Review early data collection.
- Evaluate search query effectiveness.
- Confirm final collection strategy.

Discussion/Decisions:

- Reviewed initial YouTube search results.
- Discussed issues with some game search terms returning irrelevant content.
- Refined search queries to improve relevance and increase video coverage.
- Confirmed final data collection approach and filtering criteria.

Action Items:

- Ethan: rerun collection and preprocessing pipeline.
- Roopkotha: review collected outputs and assist with dataset validation.

Meeting 3 – Week 11 (Analysis Review)

Attendees: Ethan Bito (s4102812), Roopkotha Chakrabarti (s4054393)

Agenda:

- Review sentiment analysis outputs.
- Review network metrics and visualisations.
- Discuss interpretation of findings.

Discussion/Decisions:

- Reviewed sentiment distributions across both communities.
- Compared network density, centrality, and community detection results.
- Discussed key findings to include in final report.
- Agreed on final visualisations and tables for analysis section.

Action Items:

- Ethan: finalise analysis outputs and code organisation
- Roopkotha: begin drafting results and interpretation sections

Meeting 4 – Week 12 (Final Submission Review)

Attendees: Ethan Bito (s4102812), Roopkotha Chakrabarti (s4054393)

Agenda:

- Review final report draft.
- Confirm submission requirements.
- Prepare final files.

Discussion/Decisions:

- Reviewed final report structure and supporting figures.
- Confirmed required submission components (report, code, worksheet, sample data, access file).
- Checked file naming and repository packaging requirements.
- Agreed on final submission plan.

Action Items:

- Ethan: final packaging and submission checks.
- Roopkotha: final proofreading and document review.

IV. Individual Self-Reflection

Ethan Bito (s4102812)

One aspect that worked well was the division of responsibilities across data collection, sentiment analysis, network analysis, and report writing. This allowed us to progress well once the research direction was finalised. I was able to contribute more to the technical implementation.

One challenge during the project was that we spent a bit of time discussing and refining the project direction before coming to an agreement on the final topic. Initially I considered using sports-related topics as I had worked on sports data in Assignment 1 as it was easier to analyse and had good sentiment distributions. However, after discussion we agreed to analyse rhythm games which still provided good analysis.

A further challenge for me occurred in the final week where I had some technical difficulties which prevented me from accessing my student account and was unable to communicate for a few days. This created some delays in coordination and final preparation. Despite this we were able to recover and complete the assignment.

Overall, the project reinforced the importance of planning and communication. In future projects, I would aim to finalise the project direction slightly earlier and maintain backup channels of communication.

Roopkotha Chakrabarti (s4054393)

One aspect that worked well was that we were able to refine the project scope to a topic that allowed both sentiment and network analysis to be applied. Although the topic changed during planning, the final rhythm game comparison gave us a clear way to compare two similar online communities.

One challenge was that our available data source was limited to YouTube. While YouTube comments provide useful text data and reply relationships, it was not necessarily the strongest platform for studying broader fandom interactions. This is because YouTube doesn't capture wider relationships such as followers, memberships, etc. However, given the available API access and time constraints, YouTube was the most realistic option.

The project showed the importance of planning properly and choosing a data source that fits the research question. In the future, I'd aim to confirm available data sources and make sure the research question is better shaped around the strengths and limitations of the data source.